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23 December 1954
SUPERSEDING
Army 49-42
25 August 1945

MILITARY SPECIFICATION

CHARGE, EXPLOSIVE, SPRINGING
FOR ROD, EARTH, BLAST-DRIVEN, M13

1. SCOPE

1.1 This specification covers one type of explosive springing charge for blast-driven earth rod M13.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications and all drawings and specifications referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

JAN-P-106	-Packaging and Packing for Overseas Shipment-Boxes; Wood, Nailed
MIL-G-2550	-General Specification For Ammunition Except Small Arms Ammunition

STANDARDS

MILITARY

MIL-STD-129	-Marking For Shipment And Storage

DRAWINGS

ORDNANCE CORPS

P-86256	-Charge, Explosive, Springing, For Rod, Earth, Blast-Driven, M13 Assembly and Details

PUBLICATIONS

ORDNANCE CORPS

ORD-SIP-M16-304	-Charge, Explosive, Springing for Rod, Earth, Blast-Driven M13

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(Copies of specifications, standards, drawings, and publications required by the contractor in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.-The following publications, of the issue in effect on date of invitation for bids, unless otherwise stated, form a part of this specification:

CODE OF FEDERAL REGULATIONS

Title 49--Transportation: Chapter I - Interstate Commerce Commission, Parts 71 to 78 - Explosives and Other Dangerous Articles.

(Parts 71 to 78 of Title 49 of the Code of Federal Regulations are contained in a volume which comprises parts 71 to 90 of that title. Copies of this volume may be obtained upon application, accompanied by money order, coupon, or cash, to Superintendent of Documents, Government Printing Office, Washington 25, D. C. Application should cite "49 CFR 71-90").

3. REQUIREMENTS

3.1 Material.-The material used in the manufacture of the springing charge shall be as specified on the applicable drawing. Material not definitely specified shall be of the best quality normally used for the purpose in good commercial practice. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product.

3.2 The springing charge shall comply with all requirements specified on Drawing P-86256 and on all drawings referenced thereon and with all requirements specified in applicable specifications.

3.3 Assembly and loading.-After filling the charge tube assembly, the composition B shall not extend beyond the open end of the charge tube after the composition B has cooled and hardened. The weight of the composition B required to fill each tube shall be not less than 260 grams when determined as specified in 4.5. All overflow or drops of composition B shall be cleaned from the exterior surfaces of the finished charge.

3.4 Workmanship.-Workmanship shall be of the highest grade throughout and in accordance with the best standard practice for this type commodity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.-A lot shall consist of springing charges produced by one manufacturer under one contract, in one unchanged process in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.2 Sampling.-Number of samples, acceptance, and rejection criteria used for determination of lot acceptance for tests as specified herein shall be in accordance with Standard Inspection Procedure ORD-SIP-M16-304.

4.3 Inspection.-Inspection shall be as specified in Specification MIL-G-2550.

4.4 Determination of binder-lubricant content of booster-pellets.

4.4.1 Stearic acid.-The outer surface shall be removed from the sample booster pellets to obtain a composite sample of about 5 grams from the interior of the booster pellets. An accurately weighted portion of approximately 5 grams of the sample shall

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be transferred to a tared, medium-porosity, sintered-glass crucible. The sample shall be extracted with three 20-ml portions of chloroform which has been previously saturated with RDX, maintaining each portion in contact with the residue for approximately 5 minutes before applying suction. The contents of the crucible shall be stirred occasionally to break up any lumps which may persist and the sides of the crucible shall be washed down with each 20-ml portion of chloroform to insure complete removal of the stearic acid. The residue shall be aspirated until the odor of chloroform is no longer detectable, the crucible and contents placed in a drying oven at 100° to 105°C for 1/2 hour, cooled in a desiccator, and weigned. The above extraction, drying, and weighing procedure shall be repeated with individual 20-ml portions of chloroform until the loss in weight on subsequent extractions totals less than 0.003 grams. The loss in weight of the crucible plus contents, on a moisture-free basis, shall be calculated to percent stearic acid as follows to determine compliance with the requirement specified on the drawing.

$$\text{Percentage of stearic acid} = \frac{100A}{W}$$

where:

A = loss in weight of the crucible plus contents.
W = weight of sample on a moisture-free basis.

4.4.2 Graphite.-The residue remaining after the stearic acid has been removed (see 4.4.1) shall be extracted with two 20-ml portions of dimethyl formamide, maintaining each portion in contact with the residue for approximately 5 minutes with intermittent stirring before suction is applied. The crucible and contents shall be washed with a minimum of three 20-ml portions of acetone, maintaining each portion in contact with the residue for approximately 1 minute with continuous stirring before suction is applied, until the following qualitative test for RDX in the acetone washings is negative. The acetone filtrate shall be transferred to a 20-ml beaker, evaporated to dryness on a steam bath, and several drops of solution of chromotropic acid in 36N sulfuric acid added. The formation of a reddish-brown color, as compared to a pink color obtained by a blank prepared in an identical manner, indicates the presence of RDX. The residue shall be aspirated until the odor of acetone is no longer detectable, the crucible and contents shall be placed in a drying oven at 100° to 105°C for 1/2 hour, cooled in a desiccator, and weighed. The weight of the residue in the crucible shall be calculated to the percentage of graphite in the sample on a moisture-free basis as follows to determine compliance with the requirement specified on the drawing.

$$\text{Percentage of graphite} = \frac{100A}{W}$$

where:

A = increase in weight of filtering crucible.
W = weight of sample on a moisture-free basis.

4.4.3 Total lubricant-binder.-The percentage of total lubricant-binder in the sample shall be calculated by adding the percentages of stearic acid and graphite found to determine compliance with the requirement specified on the drawing.

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4.5 Determination of weight of composition B.-Sample springing charges shall be selected and inconspicuously marked for identification. The weight of Composition B used shall be determined as follows: Each sample springing charge shall be weighed before filling with Composition B. The sample springing charge shall be placed in the loading line and filled under conditions identical in all respects with those employed in filling regular springing charges. The filled springing charge shall be weighed and the weight of Composition B calculated to determine compliance with 3.3.

4.6 Determination of specific gravity of pellets.-Sample pellets which are as free of surface cavities and broken edges as possible shall be selected, and the height and diameter of each sample pellet determined to within $\pm$ .001 inch. From these measurements determine the volume of each sample pellet in cubic inches and convert this value to volume in cubic centimeters as follows:

Volume (cubic inches) x 16.39 = Volume in cubic centimeters

Weigh each sample pellet in grams to $\pm$ .01 gram and record the weights of the pellet opposite the respective volumes. Calculate the specific gravity of the sample pellet as follows to determine compliance with the requirement specified on the drawing.

$$\text{Specific gravity} = \frac{W}{V}$$

where:

W = weight of pellet in grams
V = volume of pellet in cubic centimeters

5. PREPARATION FOR DELIVERY

5.1 Packing.-The explosive sections shall be packed 40 to a box (20 sections with sleeves), in 8 rows of 5 sections each. The box shall be lined on the inside with corrugated carboard of a similar approved material, to hold the sections securely in place. The box shall comply with the requirements of Specification JAN-P-106, type 4 or 5, and shall be strapped in accordance with the appendix thereto except that only round strapping, class A, galvanized steel shall be used. The straps shall be tensioned and sealed, but not stapled.

5.2 Labeling and marking.-In addition to any special marking required by the contract of purchase order, shipments shall be marked in accordance with Code of Federal Regulations 49 CFR 71-78 and Standard MIL-STD-129.

6. NOTES

6.1 Intended use.-The springing charge covered by this specification is intended for use with the blast-driven earth rod, M13, as an explosive charge which may be inserted into the borehole after the rod has been extracted.

6.2 Ordering data.-Procurement documents should specify the title, number, and date of this specification and any desired options offered herein (see 5.1 and 5.2).

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